

# [VJC/2025/Q1]

## 1 Name your Jupyter Notebook as:

TASK1\_<your name>\_<class>\_<index number>.ipynb

A museum maintains information about its artefacts in a text file `ARTEFACTS.txt` in the following format:

```
[(artefactID, rating), ...]
```

where:

- `artefactID` is a single uppercase letter followed by 4 digits (e.g., A1234, B5678)
- `rating` is an integer between 1 and 100 inclusive representing the historical significance

Example of `ARTEFACTS.txt` contents:

```
[(A1234, 85), (B1234, 85), (C1234, 85), ...]
```

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 1.1  
Program code
```

Output:

### Task 1.1

Write a function `task1_1(artefacts_list)` that:

- takes a list of tuples `artefacts_list`
- validates that each `artefactID` starts with an uppercase letter followed by a 4-digit number between 1000 and 9999 inclusive, outputting an appropriate error message if data is invalid
- validates that each `rating` is between 1 and 100 inclusive, outputting an appropriate error message if data is invalid
- returns a list of tuples containing valid `(artefactID, rating)` pairs. [3]

Test your function with the artefacts list in `ARTEFACTS.txt` and output:

- the total number of valid artefacts read
- the first 5 artefacts in the list. [1]

# [VJC/2025/Q1 Cont ']

## Task 1.2

Write a function `task1_2(artefacts_list)` that:

- takes a list of artefact tuples `artefacts_list` as a parameter
- implements a quicksort algorithm to sort the artefacts by rating in descending order
- uses the first element as the pivot
- returns the sorted list.

[5]

Test your function with the artefacts list in `ARTEFACTS.txt`.

[1]

## Task 1.3

Write a procedure `task1_3(artefacts_list, target_rating)` that:

- takes a list of artefact tuples `artefacts_list` and the rating `target_rating` as parameters
- uses your function from **Task 1.2** to sort the artefacts by rating in descending order
- implements a recursive binary search to find all the artefacts with the `target_rating`
- returns the tuple `(artefactID, rating)` if found or `None` if not found.

[7]

Test your function with:

- `target_rating = 85`
- `target_rating = 100`

Output an appropriate message for each test case.

[2]

Save your Jupyter Notebook for Task 1.